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Roadside revegetation with native plants: Experimental seeding and transplanting of stem cuttings

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Abstract

Questions: Can we use local native plants for roadside revegetation? What cultural methods help enhance the process?

Location: Trans Canada Highway, Terra Nova National Park, Newfoundland.

Objectives: To (1) test stratification requirements for seed germination, (2) determine if germination, survival and growth of seedlings and stem cuttings of selected plants can be increased by mulching treatments and (3) identify native plants and cultural treatments useful for revegetation.

Methods: We tested seed germination of *Kalmia angustifolia*, *Iris versicolor*, *Juncus effusus*, *Eriophorum vaginatum*, *Clin-tonia borealis* and *Cornus canadensis* in a greenhouse experiment. We conducted field experiments of roadside revegetation using seeds of *K. angustifolia*, *I. versicolor*, *J. effusus* and *E. vaginatum*, as well as seedlings of *I. versicolor* and rooted stem cuttings of *Empetrum nigrum* and *Juniperus communis* after hay-mat mulch and organic matter mulch application.

Results: Stratified seeds of *K. angustifolia*, *I. versicolor*, *J. effusus* and *E. vaginatum* germinated successfully in the greenhouse, whereas *C. borealis* and *C. canadensis* seeds did not. Along roadsides, only *I. versicolor* seeds germinated. *Iris versicolor* cover increased significantly in organic matter mulch compared to hay-mat mulch and control. Transplanted *I. versicolor* seedlings had high survival in all treatments but growth was reduced in organic matter mulch. Survival and growth of stem cuttings of *E. nigrum* and *J. communis* were significantly increased on hay-mat mulch.

Application: Rooted stem cuttings of *E. nigrum* and *J. communis* planted on hay-mat mulch can be used as a practical method of roadside revegetation. These shrubs have low structure, are evergreen, and exhibit stress-tolerance properties, which make them ideal species for roadside revegetation. They are also non-palatable to wildlife. Roadside ditches can be revegetated by seeds or seedlings of *I. versicolor*. Robust roots and rhizomes of this plant may provide soil stability and dark green leaves and attractive flowers create aesthetically pleasing vegetation cover.

Keywords: *Empetrum nigrum*; *Juniperus communis*; *Iris versicolor*; Mulching; Roadside restoration; Roadside microhabitat; Seeding; Stem-cutting; Transplanting.

Introduction

Roads and highways are conspicuous and necessary structures in natural landscapes proving vital functions of transportation of goods and people. However, many studies have highlighted various adverse road effects (Forman & Alexander 1998; Parendes & Jones 2000; Gelbard & Belnap 2003). Adverse road effects are of particular concern when large highways pass through conservation areas such as national parks and ecological reserves and interfere directly with their conservation objectives (Forman & Alexander 1998; Mallik 2000; Karim & Mallik 2008). Ecologists, conservationists and road engineers recognize the need for rapid development of vegetation cover along roadsides to minimize the detrimental effects of roads on surrounding ecosystems (Hobbs & Harris 2001). Highway engineers often provide a grass cover by hydro-seeding immediately after road construction to reduce soil erosion (Andres & Jorba 2000). However, it is more desirable to revegetate roadsides with appropriate local plants than hydro-seeding with non-native species (Tormo et al. 2007).

Several authors have reported problems associated with establishing native plants in disturbed habitats (Paschke et al. 2000; Fattorini 2001). Natural colonization of native plants is unpredictable due to biological limitations such as seed dispersal, dormancy and viability and due to inhospitable properties of roadsides such as compacted soil, low soil moisture and nutrient content, high substrate temperature and chemical toxicity (Walker & Powell 1999; Karim & Mallik 2008). Revegetation of disturbed habitats with native plants is gaining support from restoration ecologists (Grantz et al. 1998; Kirmer & Mahn 2001; Motalvo et al. 2002; Tormo et al. 2007). However, this approach is rarely used by road engineers (but see Cotts et al. 1991; Petersen et al. 2004; Tyser et al. 1998).

Failure of roadside revegetation by native plants is often due to inappropriate selection of planting material and their propagation methods. Transplanted seedlings

Table 1. Eight commonly occurring native roadside plants with desirable traits selected for roadside revegetation experiments along the Trans Canada Highway of Terra Nova National Park, Newfoundland. Superscripts indicate references relating to species traits.

Species	Occurrence in roadside microhabitat(s)	Growth form	Tolerance	Reference
1. <i>Empetrum nigrum</i> ¹ 2. <i>Juniperus communis</i> ²	Side slope	Evergreen, creeping shrub ¹ Evergreen, spreading dwarf shrub ²	Saline, drought and nutrient-poor conditions ^{1,2}	Karim & Mallik (2008) ^{1,2}
3. <i>Juncus effusus</i> ¹ 4. <i>Eriophorum vaginatum</i> ² 5. <i>Iris versicolor</i> ³	Lower side slope and ditch	Tussock grasses, dense roots, cottony flowers ¹ Rhizomatous herb, green leaves and pretty blue flowers	Wet soil ^{1,2,3}	USDA, NRCS (Anon. 2001) ^{1,2,3}
6. <i>Kalmia angustifolia</i> ¹ 7. <i>Cornus canadensis</i> ²	Back slope	Evergreen, dwarf shrub, competitive, allelopathic, nice pink flowers Creeping herb, nice red fruit and foliage color in the fall ²	Organic soil ^{1,2,3}	Mallik (1987, 1993, 1994) ¹ Karim & Mallik (2008) ^{1,2,3}
8. <i>Clintonia borealis</i> ³		Rhizomatous herb, leathery dark green leaves ³		USDA, NRCS (Anon. 2001) ^{1,2,3}

or vegetative cuttings of suitable native plants can provide an effective method for relatively rapid revegetation of roadsides by avoiding the vulnerable stages of germination and seedling establishment (Paschke et al. 2000; Fattorini 2001). Mulching with organic soil and plant substrates may facilitate seed germination and seedling establishment on roadsides (Smika & Unger 1986; Ji & Unger 2001). Commercially available hay mat, jute mat or organic matter may serve the purpose of both soil conservation and favorable growing media (Rees 2002).

The main objective of our study was to develop a practical method of roadside revegetation by (1) testing the efficacy of stratification and mulching treatments on seed germination of several native plants and (2) comparing the mulching effects on survival and growth of transplanted seedlings and rooted stem cuttings along roadsides. We hypothesized that: (a) stratification of native seeds of boreal plants may enhance or be required for germination, and (b) mulching treatments would enhance seed germination and improve survival and growth of transplanted seedlings and stem cuttings of native plants along roadsides.

Material and Methods

Plant material

We selected seeds and stem cuttings of native plants naturally occurring close to roadsides of Terra Nova National Park, Newfoundland (4833° N, 5358.8° W) that possess morphological traits such as low stature facilitating traffic visibility, extensive above and below-ground systems that help erosion control, ease of propagation, tolerance to road salt, and adaptation to high and low

moisture stress in typical roadsides (Table 1). We collected ripe fruits and seeds of *Kalmia angustifolia*, *Eriophorum vaginatum*, *Clintonia borealis*, *Cornus canadensis*, *Iris versicolor*, and *Juncus effusus* in late fall (October, 2000) from five locations adjacent to the Trans Canada Highway (TCH) within the park. The seeds were air dried at room temperature (20 °C) for five days and stored in cardboard boxes for the winter in an unheated room (December-March temperatures 0 to -20 °C). Some of these seeds were used for a greenhouse experiment and the remainder used for roadside seeding experiments. We collected young shoots (1-2 year-old) of *Juniperus communis* and *Empetrum nigrum* growing naturally along the TCH roadsides for rooting of stem cuttings in the greenhouse and subsequent planting along roadsides.

Greenhouse experiment

Considering the difference in individual seed size and seed weight, we sowed separately 0.25 g seeds of *K. angustifolia* (ca. 1400 seeds), *J. effusus* (5500 seeds) and *E. vaginatum* (500 seeds) and 0.40 g seeds of *I. versicolor* (20 seeds) each in a 10 cm × 10 cm × 8 cm deep germination tray filled with sand and peat moss at 3:1 ratio. Because we had fewer seeds of *C. borealis* and *C. canadensis* and they were quite large compared to the other seeds, we planted 20 seeds of each species in one germination tray. Six replicate trays were prepared for each seed type. After planting the 36 trays (6 species × 6 replicates) were moistened with tap water at field capacity and kept chilled for seven days at 0 °C in a growth chamber. After this stratification treatment, all the trays were transferred to the Lakehead University greenhouse and maintained at 25 °C. Seeds of the same species were similarly planted as replicates

without stratification and kept in the greenhouse at 25 °C as controls. All trays were irrigated with tap water at alternate days for five weeks. Emerging seedlings were counted weekly for five weeks.

Greenhouse propagation for field experiment

Rooting of stem cuttings of *J. communis* and *E. nigrum* was conducted in Memorial University Botanical Gardens Greenhouse by planting 4-cm pieces in plastic trays (Root-trainer Spencer-Lemaire, Edmonton, Alberta) filled with a peat moss-vermiculite mixture. The cuttings were covered with plastic sheets and irrigated twice a week with tap water without any fertilizer or rooting solution. At the time of transplanting along roadsides, the rooted stem cuttings of *J. communis* and *E. nigrum* had a mean shoot height of 4 cm and 6 cm, respectively. Similar system was used to raise seedlings of *I. versicolor* from seeds. Before transplanting, *I. versicolor* seedlings had a mean height (soil surface to the tip of longest leaf) of 10 cm. Prior to transplanting on roadsides the stem cuttings and seedlings were transferred to outdoor conditions for a two-week acclimatization period.

Field experiment with seeds, stem cuttings and seedlings

We conducted the field experiments along one-year old reconstructed roadsides. Based on topography and substrate quality, the roadsides can be classified into four distinct micro-site types: shoulder, side slope, ditch, and back slope. See Karim & Mallik (2008) for

a detailed description of the microsites.

We conducted three field experiments along side slopes. First, we compared the effects of hay-mat mulch and organic matter mulch on seed germination and seedling survival of *K. angustifolia*, *I. versicolor*, *E. vaginatum* and *J. effusus*. In early July 2001 we established nine 26 m × 2 m plots with three treatments, hay-mat mulch, organic matter mulch and no mulch (control). The hay-mat mulch was made of dry wheat hay, interwoven with jute fibers. Organic matter mulch consisted of roadside topsoil that was removed during road reconstruction 10 years previously and stored in a holding area. This was spread in a 3-5 cm thick layer on the surface of reconstructed roadsides. Each of these plots was divided into nine 2 m × 2 m sub-plots, receiving three treatments: seeding, transplanting and control each with three replicates (Fig. 1). We kept buffer plots (1 m × 1 m) between the sub-plots to avoid seed mixing from adjacent sub-plots. The sub-plots were randomly assigned to *K. angustifolia*, *I. versicolor*, *J. effusus* and *E. vaginatum*.

Second, we compared the effect of organic matter and hay-mat mulch on survival and growth of transplanted seedlings of *I. versicolor*. Greenhouse grown *I. versicolor* seedlings were transplanted in 17 m × 1 m plots along the ditch following the above design. In this case the sub-plots were 1 m × 1 m. We planted 16 seedlings in each sub-plot.

Third, we tested the effects of these mulching treatments on survival, growth, and biomass of stem cuttings of *J. communis* and *E. nigrum* on side slopes. In early July 2001, we transplanted rooted stem cuttings of *E. nigrum* and *J. communis* after adding hay-mat mulch

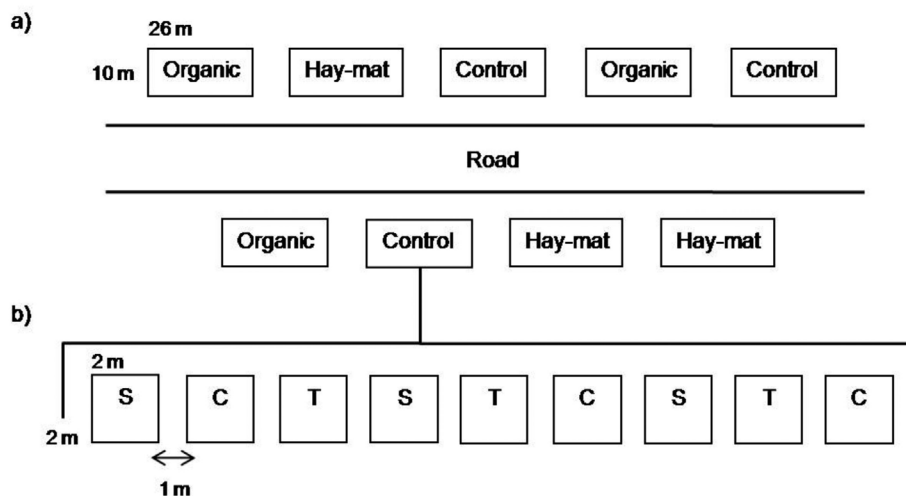


Fig. 1. Experimental layout of seeding and transplanting experiments along the newly constructed roadsides of Trans Canada Highway; **a.** Experimental plots (26 m × 2 m) receiving hay-mat mulch or organic matter mulch and control treatment; **b.** layout of sub-plots (2 m × 2 m) within each experimental plot, where S, T and C represent the seeded, transplanted and control sub-plots respectively.

and organic matter mulch separately in 17 m × 1 m plots as described above; 16 rooted stem cuttings of *E. nigrum* and *J. communis* were transplanted in each 1 m × 1 m sub-plot at 25 cm × 25 cm spacing. All experimental plots were irrigated twice a week for one month after seeding and transplanting.

Our reasons for choosing these species and microhabitats were based on field observations as well as a quantitative floristic and microhabitat substrate survey along roadsides, which showed that these plants can grow naturally in these locations (see Karim & Mallik 2008).

Data collection

Greenhouse germination was monitored weekly by counting the emerging seedlings from March to April 2001. We counted the number of emerging seedlings and their cover in 50 cm × 50 cm quadrats placed randomly in the sub-plots. We determined survival of seedlings and stem cuttings by subtracting the number of live transplants in each sub-plot in the second growing season from 16, the initial number of transplants. Initial shoot length of seedlings and transplanted stem cuttings was taken from eight randomly selected samples from each sub-plot immediately after planting. In September 2002, two growing seasons after transplanting, shoot length of the seedlings and stem cuttings was measured again from eight randomly selected seedlings and stem cuttings from each sub-plot. We measured a total of 72 seedlings or stem cuttings of each species in each treatment from the ground level to the tip. To determine root-shoot length and dry biomass we excavated 18 transplants of each species from each treatment at the end of the second growing season. The samples were dried in a convection oven at 65 °C for 24 h until constant weight.

Data analyses

We used a Kruskal-Wallis non-parametric test to compare treatment means of the greenhouse experiment since the data did not meet the requirements of a parametric test. The field experiment data met ANOVA requirements after appropriate transformation. We performed one-way ANOVA and Tukey's HSD *post hoc* tests to determine significant treatment difference (Anon. 1999).

Results

Greenhouse germination experiment

The stratification treatment significantly enhanced seed germination of *K. angustifolia*, *I. versicolor*, *J. effusus* and *E. vaginatum* ($p < 0.05$); however germination remained quite low at 2.5% and 8% in *J. effusus* and *K. angustifolia* and moderate in *E. vaginatum* and *I. versicolor* at 25% and 50%, respectively (Fig. 2a-d). Control seeds of the four species experienced very poor or no germination. *Clintonia borealis* and *C. canadensis* did not germinate. Without stratification, germination of *J. effusus*, *I. versicolor*, and *K. angustifolia* was only 0.5%, 6% and 8%, respectively and *E. vaginatum* did not germinate at all (Fig. 2a-c). The stratification treatment advanced seed germination of *K. angustifolia* by three weeks. However, at the end of the 4th week the difference was not significant compared to the control (Fig. 2b).

Roadside seeding experiments

Among the four species tested only *I. versicolor* seeds germinated on side slopes and germination was increased by 20% in organic matter mulch compared to the control ($p < 0.001$; $F = 53.7$). Germination of *I. versicolor* was lower (5%) in hay-mat mulch than control (Fig. 3). Percent cover of seedlings in the two mulching treatments showed a similar trend ($p < 0.001$; $F = 20.9$) as for germination (Fig. 3).

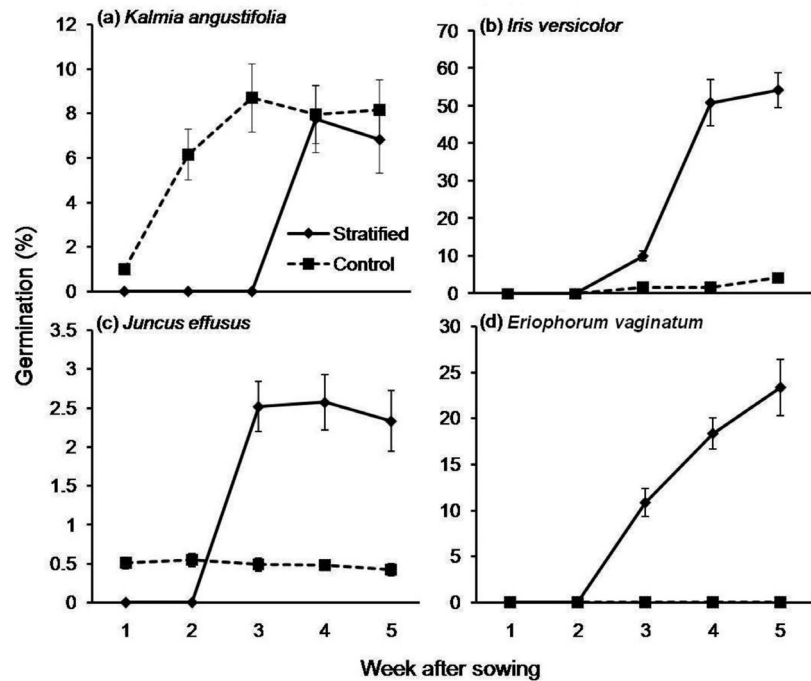
Revegetation by rooted stem cuttings

Hay-mat mulch and organic matter mulch provided higher ($p < 0.001$) survival of *E. nigrum* stem cuttings (80% and 78% respectively) compared to control (60%). The overall survival of *J. communis* cuttings was lower than *E. nigrum* in the mulching treatments. However, for *J. communis* the hay-mat mulch provided significantly ($p < 0.001$; $F = 12.5$) more survival (44%) than control and organic matter mulch (25%).

Shoot length of *E. nigrum* cuttings was almost double ($p < 0.001$; $F = 7.9$) in hay-mat mulch than in organic matter mulch (Fig. 4). Mulching had no effect on shoot length of *J. communis* cuttings ($p < 0.123$; $F = 2.1$). In general, shoot length of *J. communis* cuttings remained lower than that of *E. nigrum* in both mulching treatments.

Both *E. nigrum* and *J. communis* allocated about two-thirds of their total dry mass to above-ground parts in the mulching treatments although the total dry mass of *E. nigrum* was higher than *J. communis* in both treatments (Fig. 5a, b). There were no significant differences in root dry mass due to treatments. However, for both species shoot dry mass was lower ($p < 0.004$, 0.001 ; $F = 6.3$,

Fig. 2. Weekly cumulative average of six replicates (mean $\pm$ SE; $N = 6$ treatments) of stratified and un-stratified (control) seeds of (a) *Juncus effusus*, (b) *Kalmia angustifolia*, (c) *Iris versicolor* and (d) *Eriophorum vaginatum*. Significant treatment differences of germination at every week were determined by the Kruskal-Wallis non-parametric test.



10.3) in hay-mat mulch than in organic matter mulch and control (Fig. 4a,b). The root-shoot ratio of *E. nigrum* was higher in hay-mat mulch than control and organic matter mulch. For *J. communis*, although the highest root-shoot ratio was in hay-mat mulch, there was no difference in root-shoot biomass ratio between hay-mat mulch and organic matter mulch (Fig. 4).

Revegetation by transplanted seedlings

Mulching had no significant effect on the survival of transplanted *I. versicolor* seedlings, which remained relatively high (95%). Shoot growth of *I. versicolor*

was lower ($p < 0.001$; $F = 30.5$) in the mulching treatments than control (Fig. 5a). In control shoot length of *I. versicolor* was increased by 100% from the initial to final measurement whereas that in organic matter mulch was increased by only 25%. Total biomass of *I. versicolor* was lower ($p < 0.001$; $F = 36.2$) in hay-mat mulch and organic matter mulch (by 30 and 70%) than control (Fig. 6b). Root and shoot dry mass of the species was less in organic matter mulch than in hay-mat mulch and control. In general, *I. versicolor* seedlings allocated more biomass to roots than shoots (Fig. 6b).

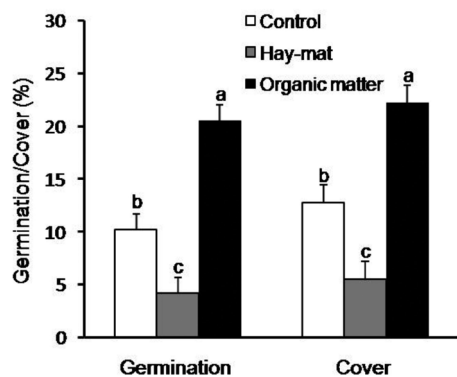


Fig. 3. Per cent germination and cover (mean $\pm$ SD; $N = 9$ sub-plots/treatment) of *Iris versicolor* one year after seeding on hay-mat mulch and organic matter mulch treatments on side slopes of Trans Canada Highway. Different letters in each category indicate significant difference at $P \leq 0.05$.

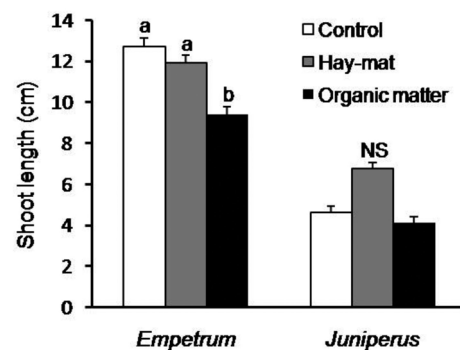


Fig. 4. Shoot length (mean $\pm$ SE; $N = 72$ cuttings per treatment) of *Empetrum nigrum* and *Juniperus communis* stem cuttings two growing seasons after transplanting on hay-mat mulch or organic matter mulch on side slopes of Trans Canada Highway. Different letters for each species indicate significant treatment differences at $P \leq 0.001$.

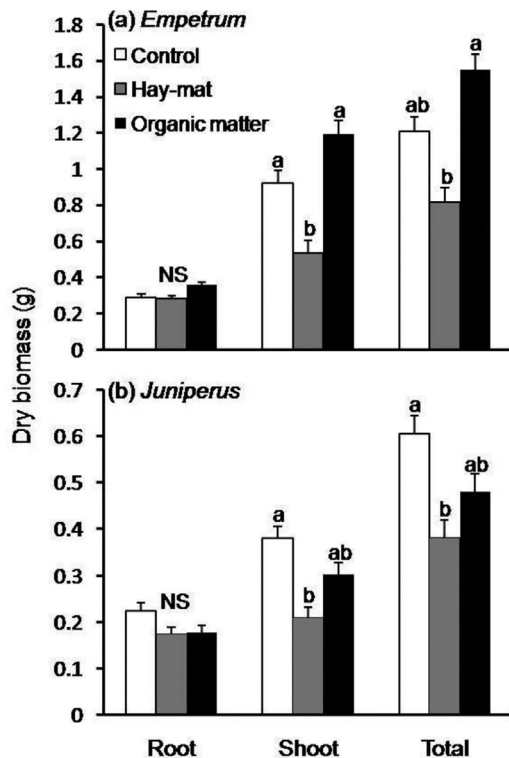


Fig. 5. Mean dry biomass ($\pm$ SD; $N = 18$ cuttings/treatment) of (a) *Empetrum nigrum* and (b) *Juniperus communis* stem cuttings two growing seasons after transplanting on hay-mat mulch or organic matter mulch on side slopes of TransCanada Highway. Different letters in each category of parameters indicate significant differences at $P \leq 0.05$ and NS = not significant.

Discussion

Seed germination in greenhouse and on roadsides

Germination of four of six native species was notably increased by greenhouse stratification treatment. When seeded on roadsides, only *I. versicolor* seeds germinated successfully. Stratification stimulates germination of many native plants of northern ecosystems (Grime 1979). The poor germination under field conditions may be attributed to inadequate stratification. Seeds of the four species that germinated in the greenhouse required both cold and moist conditions simultaneously. However, seeds used for the field experiment were air-dried and stored in an unheated room. Under natural conditions they would have been exposed to sub-zero winter temperatures as well as higher soil moisture, had they been in soil for the winter and exposed to spring snow melt. It appears that these seeds require moist and low temperature

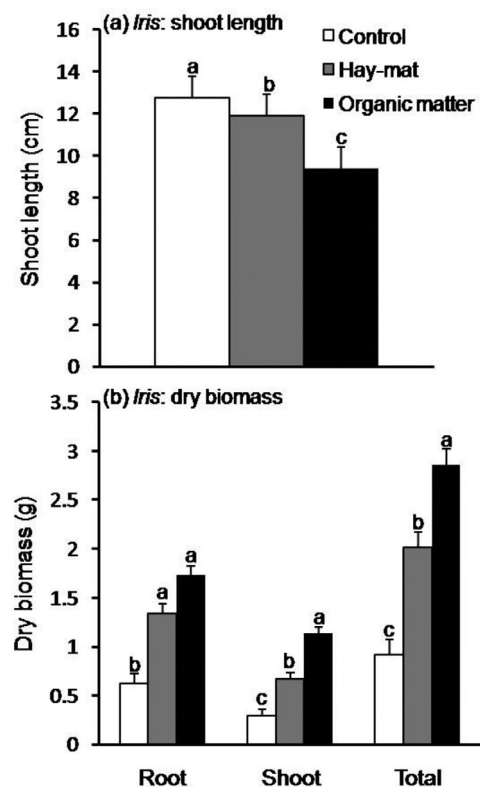


Fig. 6. a. Shoot height; b. dry biomass (mean $\pm$ SE; $N = 18$ cuttings/treatment) of *Iris versicolor* seedlings two growing seasons after transplanting on hay-mat mulch or organic matter mulch applied on the ditch microsite of TransCanada Highway. Different letters in each category of parameters indicate significant differences at $P \leq 0.05$ and NS = not significant.

conditions to break dormancy. Seeding in late fall instead of summer might be a way to satisfy the environmental requirements for germination.

Seed germination of *I. versicolor* was significantly reduced by the mulching treatments. High seed germination on dark colored 3–5 cm thick organic matter mulch can be attributed to more favorable temperature and moisture conditions than the light colored and < 1 cm thick hay-mat mulch. The cold treated seeds may require abrupt change in temperature to break dormancy as revealed by the greenhouse experiment.

Revegetation by rooted stem cuttings

Stem cuttings of *E. nigrum* survived and established better than those of *J. communis* in both mulching treatments. *E. nigrum* has an excellent capacity to survive in a wide range of ecological conditions. Excavated roots of *E. nigrum* stem cuttings showed abundant fine roots. This plant has been found to colonize in inhospitable

conditions with poor soil moisture and nutrients (Bell & Tallis 1958). In contrast, although *J. communis* is a naturally abundant plant in nutrient-poor habitats, it may not be as stress-tolerant as *E. nigrum*, because of its fewer fine roots (Houle & Babeux 1994).

Hay-mat mulch significantly increased the survival of both *E. nigrum* and *J. communis* cuttings. However, it caused significant reduction in shoot dry mass compared to organic matter mulch and control. We attribute this to inadequate adventitious root development in the transplants. We observed that poorly rooted plugs of some stem cuttings were brought to the surface by spring frost. Inadequate root development in moisture stressed condition of the side slope may have made them prone to frost heaving damage and poor shoot biomass on hay-mat mulch. Conversely, the relatively thicker and darker organic matter mulch may have provided better soil moisture and temperature conditions for root and shoot growth of the stem cuttings.

In organic matter mulch, the stem cuttings of *E. nigrum* survived better than *J. communis*. Organic matter mulch may have provided adequate moisture and nutrients leading to a successful establishment of *E. nigrum*. Poor survival of *J. communis* cuttings in organic matter mulch might have caused from its toxicity (deicing salt, lead etc. since it was collected from the roadside topsoil) that damaged the adventitious roots (Power et al. 1978; Thompson & Rutter 1986; Spencer & Port 1988). The salt and lead toxicity might not affect *E. nigrum* cuttings as much as *J. communis*, since the former is known to have extensive ericoid mycorrhizae with high stress tolerating properties (Monni et al. 2000, 2001). Further investigations are needed to determine moisture and chemical toxicity of the mulch and tolerance of the two species.

Shoot length of *E. nigrum* and *J. communis* was higher in hay-mat mulch than in organic matter mulch but their shoot biomass was lower in hay-mat mulch than organic matter mulch. Organic matter mulch may have supplied sufficient moisture and nutrients enabling the transplants to allocate more biomass for lateral growth. Plants grown in stress conditions tend to be more conservative in investing their limited energy for height growth (Grime 1979). Low root-shoot ratios of stem cuttings of *E. nigrum* and *J. communis* in organic matter mulch and control may also suggest higher allocation of energy to lateral shoot increment. In contrast, higher root-shoot ratio of the cuttings in hay-mat mulch may suggest that these cuttings allocated more energy to root elongation in search of moisture and nutrients (Tilman 1985).

Revegetation by transplanted seedlings

Mulching did not have any effect on survival of transplanted *I. versicolor* seedlings. This species may be naturally resilient in severely disturbed and stressed habitats because of its successful vegetative reproduction from stem bases, rhizomes and fibrous roots (Kron & Stewart 1994). Seedlings of *I. versicolor* allocated more biomass to below ground than above ground parts. Contrary to expectations, the organic matter mulch decreased seedling growth. It is possible that allelochemicals and pollutants in organic matter might have caused this growth inhibition (Thompson & Rutter 1986; Mallik 1987). This needs further investigation.

Conclusions and Management implications

Several conclusions can be drawn from this study:

1. Direct seeding of native plants for roadside revegetation requires stratification of seeds prior to sowing. In ditch microhabitats seeding with *I. versicolor* can develop rapid vegetation cover and help soil stabilization.

2. Both seedling and stem cuttings can be used to revegetate newly constructed roadsides. Hay-mat mulch can be used where the side slopes are steep and vulnerable to surface run off. However, spring frost heaving can be a serious problem. One possible solution to this might be to plant the stem cuttings with extensive adventitious roots. This can be accomplished in two ways; (1) by planting larger stem cuttings in early summer allowing sufficient time for root development and (2) by planting bare rooted stem cuttings. The later can be produced by a two-stage method; first, the stem cuttings can be rooted in conventional multi-pots and then in nursery beds to allow free-form root development. Planting bare root stem cuttings in early summer will ensure sufficient root development and minimize frost heaving damage. The use of roadside organic matter as a mulch is not recommended because it may contain toxic elements affecting survival and root growth of transplants.

3. Further studies are required to determine the most appropriate culture methods for vegetative propagation of other desirable native plants by greenhouse and roadside experiments.

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